

ELITE MATHEMATICS

Pearson Edexcel International GCSE

# Modular Unit 2

## Adaptive Classified

OL-T07 · Compound Interest and Depreciation

### With Answers

Questions with full worked solutions

■ 34 pages

# FAMILY MAP

## Compound Interest Depreciation

|            |                                                    |                |
|------------|----------------------------------------------------|----------------|
| <b>F01</b> | <b>Accumulate compound growth</b>                  | 8 Q   30 marks |
| <b>F02</b> | <b>Accumulate depreciation</b>                     | 2 Q   6 marks  |
| <b>F03</b> | <b>Compare compound and mixed investment paths</b> | 3 Q   11 marks |
| <b>F04</b> | <b>Recover a missing compound parameter</b>        | 2 Q   6 marks  |

**Question 15**

4MA1-2H Q9

ORIGINAL QUESTION

3 marks

- 9 Giovanni invests 4500 koruna in a savings account for 4 years.  
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.

Give your answer correct to the nearest koruna.

..... koruna

**Worked Solution 15**

Compound interest

MODULAR UNIT 2

3 marks

**Multiplier**

$$4500(1.024)^4 = 4947.802 \dots$$

**Final answer**

4948 koruna

**Question 26**

4MA1-2H Q7 whole

ORIGINAL QUESTION UNIT 2

3 marks

- 7 Hermione buys a boat for \$26 800  
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.  
Give your answer correct to the nearest dollar.

\$.....

**Worked solution 26**

Compound depreciation

MODULAR UNIT 2

3 marks

**whole – Compound depreciation**

Use the depreciation multiplier

Each year multiplies the value by  $1 - 0.08 = 0.92$ .

Apply it for three years

$$V = 26800(0.92)^3 = 20868.8384.$$

Round to dollars

$$V \approx \$20\,869.$$

**Final answer**

\$20,869

**Question 22**

4MA1-2H Q8

ORIGINAL QUESTION

3 marks

- 8 Charlotte buys a painting for \$680  
The value of the painting increases by 4% each year.  
Work out the value of the painting at the end of 3 years.  
Give your answer correct to the nearest \$

\$.....

**Worked Solution 22**

Compound appreciation

MODULAR UNIT 2

3 marks

**Use the annual multiplier**

$$V = 680(1.04)^3 = 764.90752.$$

**Round as requested**

$$V \approx \$765 \text{ to the nearest dollar.}$$

**Final answer**

\$765

**Original question 23**

4MA1-2H Q9 M01

MODULAR UNIT 2

3 marks

- 9 Nisha invests 20 000 euros for 3 years in a savings account.  
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account  
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

**Worked solution 23**

Compound interest

MODULAR UNIT 2

3 marks

**M01 – Compound interest****Use the yearly multiplier**

$$1 + 0.035 = 1.035.$$

**Compound for three years**

$$20000(1.035)^3 = 22174.34 \dots$$

**Round**

$$\text{balance} = 22174 \text{ euros (nearest euro).}$$

**Final answer**

22174 euros

**Original question 07**

4MA1-1H Q6 Q6(b)

**ORIGINAL QUESTION****3 marks**

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %  
(3)

**Worked solution 07**

Chapter: Compound Interest &amp; Depreciation

MODULAR UNIT 2

3 marks

**Q6(b) – Chapter: Compound Interest & Depreciation**

Family: Accumulate depreciation • Variant: Known annual depreciation rate

**Method**

After each year, 85% of the value remains, so the multiplier is 0.85.

*Why: A 15% depreciation leaves  $100\% - 15\% = 85\%$ .***Method**After two years the remaining proportion is  $0.85^2 = 0.7225$ .*Why: Apply the same depreciation multiplier twice.***Method**The total depreciated proportion is  $1 - 0.7225 = 0.2775$ , so the depreciation is 27.75%.*Why: Subtract the remaining proportion from the original whole.***Final answer****depreciation: 27.75%**

**Original question 23**

4MA1-2H Q9 Q9

**ORIGINAL QUESTION****3 marks**

- 9 Pasha invests 50 000 dollars in a savings account for 4 years.  
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.  
Give your answer correct to the nearest dollar.

..... dollars

**Worked solution 23**

Chapter: Compound Interest &amp; Depreciation

MODULAR UNIT 2

3 marks

**Q9 – Chapter: Compound Interest & Depreciation**

Family: Accumulate compound growth • Variant: Known annual compound-interest rate

**Method**The annual multiplier is  $1 + 1.3/100 = 1.013$ .*Why: Convert the annual percentage increase to a multiplier.***Method**After 4 years the balance is  $50000(1.013)^4 = 52650.913\dots$  dollars.*Why: Compound the same multiplier for four years.***Method**

To the nearest dollar, the balance is 52651 dollars.

*Why: Round only the final balance.***Final answer****balance:** \$52,651

**Original question 08**

4MA1-1H Q11

**ORIGINAL QUESTION****3 marks**

- 11** Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and  $x\%$  for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of  $x$

Give your answer correct to one decimal place.

$x = \dots\dots\dots$

**Worked solution 08**

Chapter: Compound Interest &amp; Depreciation

MODULAR UNIT 2

3 marks

**Q11 – Chapter: Compound Interest & Depreciation**

Family: Compare compound and mixed investment paths • Variant: Changing annual percentage rates

**Method**After year 1 the value is  $200000 \times 1.018 = 203600$  yen.*Why: Apply the first-year interest rate.***Method**The two-year multiplier is  $209754/203600 = 1.015000\dots$ , so  $(1+x/100)^2 = 1.015000\dots$ *Why: Compare the final value with the value after year 1.***Method** $1+x/100 = 1.015$ , hence  $x = 1.5\%$ .*Why: Take the positive square root and convert the multiplier to a percentage.***Final answer**rate:  $x=1.5\%$

- 11** Max invests \$6000 in a savings account for 3 years.  
The account pays compound interest at a rate of 1.5% per year for the first 2 years.

The compound interest rate changes for the third year.  
At the end of 3 years, there is a total of \$6311.16 in the account.

Work out the compound interest rate for the third year.  
Give your answer correct to 1 decimal place.

..... %

## Worked solution

January 2020 | Question 11 | 3 marks

Family: Recover a missing compound parameter • Variant: Find one exceptional year's rate or the original principal

Method / working

1. After two years:  $6000(1.015)^2=6181.35$ .
  2.  $6181.35(1+r/100)=6311.16$ , so the multiplier is 1.021... .
  3.  $r=2.1\%$  to 1 decimal place.
- Final answer: 2.1%

7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

- (a) Work out the value of Chen's investment at the end of 3 years.  
Give your answer to the nearest yuan.

..... yuan  
(3)

## Worked solution

January 2021 | Question 7 | 3 marks

Family: Accumulate compound growth • Variant: Known annual compound-interest rate

Method / working

1. After three years:  $40000 \times 1.035^3 = 44348.72\dots$  yuan.

2. Round to the nearest yuan.

3. The investment is 44,349 yuan.

Final answer: 44,349 yuan

Wang invested  $P$  yuan.

The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

(b) Work out the value of  $P$ .

$P = \dots\dots\dots$   
(3)

## Worked solution

January 2021 | Question 7 | 3 marks

Family: Recover a missing compound parameter • Variant: Find one exceptional year's rate or the original principal

Method / working

1. A 6.5% decrease leaves 93.5%, so  $0.935P=30481$ .

2.  $P=30481/0.935=32600$ .

3. Therefore  $P=32600$  yuan.

Final answer:  $P=32,600$  yuan

7 Ali and Badia each have 25 000 dollars to invest.

| Cyclone Bank                                                            | Tornado Bank                                                                    |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Invest 25 000 dollars<br>4.5% compound interest per year<br>for 3 years | Invest 25 000 dollars<br>Receive 1150 dollars interest each year<br>for 3 years |

Ali invests in the Cyclone Bank for 3 years.

Badia invests in the Tornado Bank for 3 years.

By the end of the 3 years, Ali will have received more interest than Badia.

How much more?

Show your working clearly.

Give your answer correct to the nearest dollar.

..... dollars

# Worked solution

November 2021 | Question 7 | 4 marks

Family: Compare compound and mixed investment paths | Variant: Compound versus simple or alternative return

## METHOD / WORKING

|                                             |
|---------------------------------------------|
| 1. Ali's interest from Cyclone Bank is      |
| 2. $25000(1.045)^3 - 25000 = 3529.153\dots$ |
| 3. Badia's interest from Tornado Bank is    |
| 4. $1150 \times 3 = 3450$                   |
| 5. Difference:                              |
| 6. $3529.153\dots - 3450 = 79.153\dots$     |
| 7. Correct to the nearest dollar,           |
| 8. $79.153\dots \approx 79$                 |
| 9. Answer: \$79                             |

**FINAL ANSWER**

\ \$79

- 13** Jan invests \$8000 in a savings account.  
The account pays compound interest at a rate of  $x\%$  per year.  
At the end of 6 years, there is a total of \$8877.62 in the account.  
Work out the value of  $x$ .  
Give your answer correct to 2 decimal places.

$x = \dots\dots\dots$

**(Total for Question 13 is 3 marks)**

## Answer reference | May-Nov 2020 | Question 13

Family: Accumulate compound growth • Variant: Known annual compound-interest rate

|    |                                                                                                                                                                                                                                     |      |   |                      |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---|----------------------|
| 13 | $8000 \times \left( \frac{100+x}{100} \right)^6 = 8877.62$ oe <b>or</b><br>$8000 \times \left( 1 + \frac{x}{100} \right)^6 = 8877.62$ oe <b>or</b><br>$8000 \times (1+x\%)^6 = 8877.62$ <b>or</b><br>$8000 \times y^6 = 8877.62$ oe |      | 3 | M1                   |
|    | $\left( \frac{8877.62}{8000} \right)^{\frac{1}{6}} (=1.0175\dots)$ or<br>$(1.1097\dots)^{\frac{1}{6}} (=1.0175\dots)$                                                                                                               |      |   | M1                   |
|    |                                                                                                                                                                                                                                     | 1.75 |   | A1                   |
|    |                                                                                                                                                                                                                                     |      |   | <b>Total 3 marks</b> |

## Worked solution

May-Nov 2020 | Question 13 | 3 marks

Family: Accumulate compound growth • Variant: Known annual compound-interest rate

Method

Let the compound interest rate be  $x\%$ .

$$8000(1+(x)/(100))^6=8877.62$$

$$1+(x)/(100)=\sqrt[6]{(8877.62)/(8000)}=1.017500\dots$$

$$x=1.7500\dots$$

Correct to 2 decimal places,

$$x=1.75$$

Answer:  $x=1.75$

- 5 Himari's annual salary is 3 130 000 Japanese Yen (JPY).  
She gets a salary increase of 4%

(a) Work out Himari's salary after this increase.

.....JPY  
(3)

Kaito bought a car.

The value of the car when Kaito bought it was 750 000 JPY.

At the end of each year, the value of his car had depreciated by 15%

- (b) Work out the value of Kaito's car at the end of 3 years.  
Give your answer correct to the nearest JPY.

.....JPY  
(3)

(Total for Question 5 is 6 marks)

## Answer reference | May-Nov 2020 | Question 5

Family: Accumulate compound growth • Variant: Known annual asset appreciation

|   |   |                                                                                      |                          |   |                                                                                                                                                                                            |
|---|---|--------------------------------------------------------------------------------------|--------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | a | $1.04 \times 3\,130\,000$ oe                                                         |                          |   | M2 complete method to increase salary by 4%<br>M1 for $0.04 \times 3\,130\,000$ oe<br>(= 125 200)                                                                                          |
|   |   |                                                                                      | 3 255 200                | 3 | A1                                                                                                                                                                                         |
|   | b | for $0.15 \times 750\,000$ oe (=112 500)<br>or $0.85 \times 750\,000$ oe (=637 500)  | OR                       |   | M1 For method to find depreciation for 1 year or value after 1 year                                                                                                                        |
|   |   | $0.85 \times "637\,500"$ oe (= 541 875)<br>$0.85 \times "541\,875"$ oe(= 460 593.75) | $750\,000 \times 0.85^3$ |   | M1 for completing method                                                                                                                                                                   |
|   |   |                                                                                      | 460 594                  | 3 | A1 accept 460 593 – 460 594                                                                                                                                                                |
|   |   |                                                                                      |                          |   | SC: if no other marks gained award M1 for<br>$0.55 \times 750\,000$ oe (= 412 500) or<br>$0.45 \times 750\,000$ oe (= 337 500)<br><br>accept $(1 - 0.15)$ as equivalent to 0.85 throughout |
|   |   |                                                                                      |                          |   | <b>Total 6 marks</b>                                                                                                                                                                       |

## Worked solution

May-Nov 2020 | Question 5 | 6 marks

Family: Accumulate compound growth • Variant: Known annual asset appreciation

Official final mark-scheme pages are embedded immediately after this question.

8 Matteo is going to invest 5000 Swiss francs for two years.

He can invest his money in Bank **G** or in Bank **H**.

Bank **G**  
1.6% per year  
compound interest

Bank **H**  
2.9% interest added after  
two years

The total amount of interest Matteo would receive at the end of two years from Bank **G** is more than the amount of interest Matteo would receive at the end of two years from Bank **H**.

How much more?

..... Swiss francs

**(Total for Question 8 is 4 marks)**

## Answer reference | January 2023 | Question 8

Family: Compare compound and mixed investment paths • Variant: Compound versus simple or alternative return

| Question | Working                                                                                                                                                                                              |                                                | Answer | Mark | Notes         |  |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------|------|---------------|--|
| 8        | $\frac{2.9}{100} \times 5000 (= 145)$ oe or $1.029 \times 5000 (= 5145)$ oe or<br><br>$1.029^2 \times 5000 (= 5294....)$ oe or $0.058 \times 5000 (= 290)$ oe<br><br>or $1.058 \times 5000 (= 5290)$ |                                                |        |      | M1 Bank H     |  |
|          | $5000 \times 0.016$ oe (= 80) oe<br><br>or $5000 \times 1.016$ oe (= 5080) oe<br><br>or $5000 \times 0.032 (= 160)$ oe<br><br>or $5000 \times 1.032 (= 5160)$ oe                                     | M2 for<br>$5000 \times 1.016^2$<br>(= 5161.28) |        | 4    | M1 Bank G     |  |
|          | $(80 + 5000) \times 0.016 (= 81.28)$ oe<br><br>or $5080 \times 1.016 (= 5161.28)$ oe                                                                                                                 |                                                |        |      | M1 Bank G     |  |
|          | Correct answer scores full marks (unless from obvious incorrect working)                                                                                                                             |                                                | 16.28  |      | A1            |  |
|          |                                                                                                                                                                                                      |                                                |        |      | Total 4 marks |  |

## Worked solution

January 2023 | Question 8 | 4 marks

Family: Compare compound and mixed investment paths • Variant: Compound versus simple or alternative return

Official final mark-scheme pages are embedded immediately after this question.

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %  
(3)

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs  
(3)

(Total for Question 5 is 6 marks)

## Worked solution

November 2025 | Question 5 | 6 marks

Family: Accumulate compound growth • Variant: Known annual asset appreciation

Method

For part (a), the increase is

$$87 - 75 = 12$$

Percentage increase:

$$(12)/(75) \times 100 = 16$$

For part (b), a depreciation of 20% gives multiplier

$$0.80$$

After 3 years,

$$960(0.80)^3 = 491.52$$

Answer: (a) 16%, (b) 491.52 Swiss francs

PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION